

DS	Session	Tree 2	Question Information	Level 1 Challenge 75
Question description A beautiful <i>code</i> of a tree of n nodes is a sequence of $n-2$ integers that uniquely specifies the structure of the tree. The code is constructed as follows: As long as there are at least three nodes left, find a leaf with the smallest label, add the label of its only neighbour to the code, and remove the leaf from the tree. Given a beautiful code of a tree, your task is to construct the original tree. Constraints • $3 \leq n \leq 2 \cdot 10^5$ • $1 \leq a, b \leq n$ Input The first input line contains an integer n: the number of nodes. The nodes are numbered $1, 2, \dots, n$. The second line contains $n-2$ integers: the beautiful code. Output Print $n-1$ lines describing the edges of the tree. Each line has to contain two integers a and b: there is an edge between nodes a and b. You can print the edges in any order.				

✓ Logical Test Cases

Test Case 1
INPUT (STDIN)
5
2 2 4
EXPECTED OUTPUT
1 2
3 2
2 4
4 5

Test Case 2
INPUT (STDIN)
5
3 1 2
EXPECTED OUTPUT
4 3
3 1
1 2
2 5

✓ Mandatory Test Cases

Test Case 1
KEYWORD
q.pop();

✓ Complexity Test Cases

Test Case 1
CYCLOMATIC COMPLEXITY
3

Test Case 2
TOKEN COUNT
300

Test Case 3
NLOC
34